

Multiple Choice (10 marks)

1. Which of the following genes conveys susceptibility for polygenic Alzheimer disease?
 - (a) APP
 - (b) PS 1
 - (c) PS 2
 - (d) APOE
2. The normal human chromosome diploid number is:
 - (a) 23
 - (b) 24
 - (c) 46
 - (d) 48
3. Most viral vaccines are produced by which of the following techniques?
 - (a) Traditional cell culture
 - (b) Molecular biology
 - (c) Animal tissues in vitro
 - (d) VLPs (virus like particles)
4. The highest risk of HIV infection among drug users in the U.S. is in:
 - (a) The east coast
 - (b) The central United States (Kansas, Missouri etc.)
 - (c) Southern California
 - (d) The Pacific Northwest (Washington, Oregon, Idaho)
5. Free radicals are produced by
 - (a) Normal metabolism
 - (b) Excess consumption of Vitamins A and C
 - (c) Lack of sunlight
 - (d) A weakened immune system

True / False (10 marks)

Answer True or False

6. True or False: The dorsal roots of all spinal nerves contain sensory neuronal processes that transmit sensory information to the central nervous system.
7. True or False: The gender gap shows no significant differences in risky behaviors and smoking rates among males and females.
8. True or False: The trachea lies superficial to the oesophagus.
9. True or False: When palpating the anterior midline of the neck, the cricoid cartilage is felt before the thyroid cartilage and hyoid bone.
10. True or False: As of 2019, African American women are more likely to remain at home alone compared to other demographic groups.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of tar in tobacco smoke and its contribution to cancer development. Analyze the mechanisms by which tar affects cellular processes and promotes tumorigenesis.
12. Discuss the role of cellular scaffolding in the nucleus and cytoplasm during the assembly stage of viral replication, highlighting its importance and implications for viral lifecycle.

End of Paper